

# MAT 528 HW03

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**Problem 1** (Munkres 19.6). Let  $\mathbf{x}_1, \mathbf{x}_2, \dots$  be a sequence of the points of the product space  $\prod_{\alpha \in I} X_\alpha$ . Show that this sequence converges to the point  $\mathbf{x}$  if and only if the sequence  $\pi_\alpha(\mathbf{x}_1), \pi_\alpha(\mathbf{x}_2), \dots$  converges to  $\pi_\alpha(\mathbf{x})$  for each  $\alpha$ . Is this fact true if one uses the box topology instead of the product topology?

*Solution.* Let  $X = \prod_{\alpha \in I} X_\alpha$ . Suppose that  $\mathbf{x}_1, \mathbf{x}_2, \dots \in X$  converges to  $\mathbf{x}$  and fix an arbitrary  $\alpha \in I$ . If  $V \subseteq X_\alpha$  is an open set containing  $\pi_\alpha(\mathbf{x})$  then  $U = \pi_\alpha^{-1}(V) \subseteq X$  is an open set (by definition of the product topology) that contains  $\mathbf{x}$ . Then  $U$  contains cofinitely many of the  $\mathbf{x}_j$ , so  $\pi_\alpha(U) = V$  contains cofinitely many of the  $\pi_\alpha(\mathbf{x}_j)$ . Since  $V$  was an arbitrary open set, this means that any open set  $V \subseteq X_\alpha$  containing  $\pi_\alpha(\mathbf{x})$  contains cofinitely many points in the sequence  $(\pi_\alpha(\mathbf{x}_j))_{j \in \mathbb{N}}$ , which is what it means for this sequence to converge to  $\pi_\alpha(\mathbf{x})$ .

Conversely suppose that for some sequence  $\mathbf{x}_1, \mathbf{x}_2, \dots \in X$ , the sequence  $\pi_\alpha(\mathbf{x}_1), \pi_\alpha(\mathbf{x}_2), \dots \in X_\alpha$  converges to  $\pi_\alpha(\mathbf{x})$  for each  $\alpha \in I$ . Let  $V \subseteq X$  be a set of the form

$$\pi_{\alpha_1}^{-1}(V_1) \cap \dots \cap \pi_{\alpha_n}^{-1}(V_n)$$

for some indices  $\alpha_1, \dots, \alpha_n$  and open sets  $V_i \subseteq X_{\alpha_i}$ . For each  $i = 1, \dots, n$ ,  $V_i$  contains cofinitely many  $\pi_{\alpha_i}(\mathbf{x}_j)$  because  $V_i \subseteq X_{\alpha_i}$  is an open set containing  $\pi_{\alpha_i}(\mathbf{x})$ , hence the pre-images of each  $V_i$  contains cofinitely many  $\mathbf{x}_j$ . Equivalently,  $\pi_{\alpha_i}(V_i)^c$  contains finitely many points  $\pi_{\alpha_i}(\mathbf{x}_j)$ , so their union

$$\pi_{\alpha_1}(V_1)^c \cup \dots \cup \pi_{\alpha_n}(V_n)^c$$

also contains finitely many points of the sequence  $(\mathbf{x}_j)_{j \in \mathbb{N}}$ . Therefore the complement of this union,

$$\pi_{\alpha_1}(V_1) \cap \dots \cap \pi_{\alpha_n}(V_n),$$

contains cofinitely many of the  $\mathbf{x}_j$ . Sets of this form (intersections of finitely pre-images of open sets in some  $X_\alpha$ s) are a basis for the product topology on  $X$ , so any open set  $U$  with  $\mathbf{x}$  as an element contains such a set  $B$ ; since only finitely many of the  $\mathbf{x}_j$  are outside  $B$ , only finitely many of the  $\mathbf{x}_j$  can be outside the bigger set  $U$ . Since  $U$  is arbitrary we have shown that any open set  $U$  containing  $\mathbf{x}$  also contains cofinitely many  $\mathbf{x}_j$ , hence  $\mathbf{x}_1, \mathbf{x}_2, \dots$  converges to  $\mathbf{x}$ .

The problem statement does not hold in the box topology. Consider the sequence  $\mathbf{x}_n = (1/n, 1/n, \dots) \in \mathbb{R}^\omega$  (where of course  $\mathbb{R}^\omega$  has the box topology inherited from giving each copy of  $\mathbb{R}$  the standard topology). Then, projecting this sequence into any of the copies of  $\mathbb{R}$  gives a sequence converging to zero, so if the problem statement were to hold one should expect the  $\mathbf{x}_n$  to converge to  $(0, 0, \dots)$ . However, the neighborhood  $\prod_{n=1}^\infty (-1/n, 1/n)$  of  $(0, 0, \dots)$  does not contain any of the  $\mathbf{x}_n$  at all, so the  $\mathbf{x}_n$  don't converge to  $(0, 0, \dots)$ .  $\square$

**Problem 2** (Munkres 20.3). Let  $X$  be a metric space with metric  $d$ .

- (a) Show that  $d: X \times X \rightarrow \mathbb{R}$  is continuous.
- (b) Let  $X'$  denote a space with the same underlying set as  $X$ . Show that if  $d: X' \times X' \rightarrow \mathbb{R}$  is continuous, then the topology of  $X'$  is finer than the topology of  $X$ .

*Solution.* (a) Since the topology of  $X$  is generated by the basis of open balls  $B_r(a) = \{x \in X : d(a, x) < r\}$  for  $a \in X$ ,  $r \in (0, \infty)$ , the topology of  $X \times X$  has basis

$$\mathcal{B} = \{B_{r_1}(a_1) \times B_{r_2}(a_2) : r_1, r_2 \in (0, \infty), a_1, a_2 \in X\}.$$

To show that  $d$  is continuous, it suffices to verify that for every open interval  $(p, q) \subseteq \mathbb{R}$  with  $p < q$ ,  $d^{-1}((p, q))$  is open in  $X \times X$ ; it is sufficient because the open intervals are a basis for the topology of  $\mathbb{R}$ .

To show that  $d^{-1}((p, q))$  is open in  $X \times X$ , we want to show that for any element of  $(x, y) \in d^{-1}((p, q))$  there is an element  $B$  of  $\mathcal{B}$  such that  $(x, y) \in B \subseteq d^{-1}((p, q))$ . Let  $r = d(x, y)$  and define  $\delta = \frac{\min(r-p, q-r)}{2}$ . Then I claim that if  $(x', y') \in B = B_\delta(x) \times B_\delta(y)$ ,  $d(x', y') \in (p, q)$ , or equivalently  $(x', y') \in d^{-1}((p, q))$ . By a few applications of the triangle inequality, we get the bounds

$$\begin{aligned} d(x', y') &\leq d(x', x) + d(x, y') \leq d(x', x) + d(x, y) + d(y, y') < r + 2\delta \leq r + 2 \cdot \frac{q-r}{2} = q \\ d(x', y') &\geq d(x, y') - d(x, x') \geq d(x, y) - d(y, y') - d(x, x') > r - 2\delta \geq r - 2 \cdot \frac{r-p}{2} = p. \end{aligned}$$

Therefore any  $(x', y') \in B$  is in the pre-image of  $(p, q)$ , so  $B \subseteq d^{-1}((p, q))$ . That is, for every  $x \in d^{-1}((p, q))$ , we have found a basis element containing  $x$  and contained in  $d^{-1}((p, q))$ , which implies that  $d^{-1}((p, q))$  is open in  $X \times X$  for all open intervals  $(p, q)$ , so  $d: X \times X \rightarrow \mathbb{R}$  is continuous.

(b)  $d$  is continuous in each coordinate so for any fixed  $a \in X'$ ,  $d_a = d|_{\{a\} \times X'}: X' \rightarrow \mathbb{R}$  is continuous. Then for any  $r \in \mathbb{R}_{>0}$ , the pre-image  $d_a^{-1}((-r, r))$  is  $\{(a, x) : d(a, x) < r\}$ .

**Lemma 2.1.**  $U \subseteq X'$  is open if and only if  $\{a\} \times U \subseteq \{a\} \times X'$  is open (where  $\{a\} \times X'$  has the subspace topology).

*Proof.* Suppose that  $U \subseteq X'$  is open. Then  $\{a\} \times U = (X' \times U) \cap (\{a\} \times X')$  is the intersection of an open subset  $X' \times U \subseteq X' \times X'$  (because products of open sets are open in the product topology on  $X' \times X'$ ) with  $\{a\} \times X'$ , which means it's open in the subspace topology on  $\{a\} \times X'$ .

Conversely suppose  $\{a\} \times U$  is open in  $\{a\} \times X'$ . Then there is an open set  $V \subseteq X' \times X'$  such that  $V \cap (\{a\} \times X') = \{a\} \times U$ . Then,  $V \cap (X' \times U)$  is also open in  $X'$  and the projection  $U' = \pi_2(V \cap (X' \times U))$  in the second coordinate is an open set because  $\pi_2$  is an open map due to HW1 #5. On one hand  $U'$  is contained in  $U$  because it is a subset of  $\pi_2(X' \times U) = U$ , and on the other hand it contains  $U$  because  $V$  contains every point of the form  $(a, u)$  for some  $u \in U$ , so the open set  $U'$  is equal to  $U$ .  $\square$

By the lemma, the map  $p: X' \rightarrow \{a\} \times X'$  given by  $p(x) = (a, x)$  is a homeomorphism: it is bijective because it has an inverse  $p^{-1}(a, x) = x$ , it is continuous because the pre-image under  $p$  of an open  $\{a\} \times U \subseteq \{a\} \times X'$  is the open (by the Lemma) set  $U$ , and its inverse is continuous because the pre-image of an open  $U$  under  $p^{-1}$  is  $\{a\} \times U$  which is again open by the Lemma.

Now we are set to solve the problem: the set

$$p(d_a^{-1}((-r, r))) = p(\{(a, x) \in \{a\} \times X' : d(a, x) < r\}) = \{x \in X' : d(a, x) < r\} = B_r(a)$$

is open in  $X'$  by the work above, which means that every basis element of  $X$  (the open balls of  $d$ ) is open in  $X'$ , so the topology generated by the basis of  $X$ , i.e. the topology of  $X$ , is coarser than that of  $X'$  because the generated topology is the minimal one containing the topology of  $X$ .  $\square$

**Problem 3** (Munkres 20.10). Let  $X$  denote the subset of  $\mathbb{R}^\omega$  consisting of all sequences  $(x_1, x_2, \dots)$  such that  $\sum x_i^2$  converges.

- (a) Show that if  $\mathbf{x}, \mathbf{y} \in X$ , then  $\sum |x_i y_i|$  converges.
- (b) Let  $c \in \mathbb{R}$ . Show that if  $\mathbf{x}, \mathbf{y} \in X$ , then so are  $\mathbf{x} + \mathbf{y}$  and  $c\mathbf{x}$ .
- (c) Show that

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^{\infty} (x_i - y_i)^2}$$

is a well-defined metric on  $X$ .

*Solution.* (a) We know that  $\sum_{i=1}^{\infty} x_i^2$  and  $\sum_{i=1}^{\infty} y_i^2$  both converge absolutely because all their addends are nonnegative, so the series  $\sum_{i=1}^{\infty} (x_i^2 + y_i^2)$  is also (absolutely) convergent. We can then bound  $x_i^2 + y_i^2 \geq 2|x_i y_i| \geq |x_i y_i|$ , so since  $|x_i y_i|$  is also nonnegative we can conclude that

$$\sum_{i=1}^{\infty} |x_i y_i| \leq \sum_{i=1}^{\infty} (x_i^2 + y_i^2) < \infty$$

is convergent.

- (b) We need to show that  $\sum_{i=1}^{\infty} (x_i + y_i)^2$  converges. This follows from rearranging the sum as

$$\sum_{i=1}^{\infty} (x_i + y_i)^2 = \sum_{i=1}^{\infty} x_i^2 + \sum_{i=1}^{\infty} 2x_i y_i + \sum_{i=1}^{\infty} y_i^2$$

where the first and last sum converge by  $\mathbf{x}, \mathbf{y} \in X$  and the middle sum converges because  $\sum_{i=1}^{\infty} |x_i y_i|$  converges, and of course doubling all the addends of a convergent series yields another convergent series. This proves  $\mathbf{x} + \mathbf{y} \in X$ .

Then, given  $c \in \mathbb{R}$  and  $\mathbf{x} \in X$ , we find that

$$\sum_{i=1}^{\infty} (cx_i)^2 = c^2 \sum_{i=1}^{\infty} x_i^2$$

is also convergent so  $c\mathbf{x}$  is in  $X$  as well.

- (c) Note that

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^{\infty} x_i y_i$$

converges if  $\mathbf{x}, \mathbf{y} \in X$  by part (a) and is an inner product on  $X$ , and the norm of  $\mathbf{x}$  induced by this inner product is  $\sqrt{\mathbf{x} \cdot \mathbf{x}} = d(\mathbf{x}, \mathbf{0})$ . By the Cauchy-Schwarz inequality,  $|\mathbf{x} \cdot \mathbf{y}| \leq d(\mathbf{x}, \mathbf{0})d(\mathbf{y}, \mathbf{0})$ , so

$$\begin{aligned} d(\mathbf{x}, \mathbf{y})^2 &= (\mathbf{x} - \mathbf{y}) \cdot (\mathbf{x} - \mathbf{y}) \\ &= d(\mathbf{x}, \mathbf{0})^2 + d(\mathbf{y}, \mathbf{0})^2 - 2\mathbf{x} \cdot \mathbf{y} \\ &\leq d(\mathbf{x}, \mathbf{0})^2 + d(\mathbf{y}, \mathbf{0})^2 + 2d(\mathbf{x}, \mathbf{0})d(\mathbf{y}, \mathbf{0}) \\ &= (d(\mathbf{x}, \mathbf{0}) + d(\mathbf{y}, \mathbf{0}))^2. \end{aligned}$$

Since the quantities being squared are both nonnegative we can safely take the square root to recover the triangle inequality when the point in the middle is zero; this is sufficient because

$$d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y}) = d(\mathbf{x} - \mathbf{z}, \mathbf{0}) + d(\mathbf{0}, \mathbf{y} - \mathbf{z}) \geq d(\mathbf{x} - \mathbf{z}, \mathbf{y} - \mathbf{z}) = d(\mathbf{x}, \mathbf{y}).$$

□

**Problem 4** (Munkres 21.2). Let  $X$  and  $Y$  be metric spaces with metrics  $d_X$  and  $d_Y$ , respectively. Let  $f: X \rightarrow Y$  have the property that for every pair of points  $x_1, x_2 \in X$ ,

$$d_X(x_1, x_2) = d_Y(f(x_1), f(x_2)).$$

Show that  $f$  is an embedding. It is called an **isometric embedding** of  $X$  in  $Y$ .

*Solution.*  $f$  is injective (so  $f: X \rightarrow f(X)$  is bijective) because

$$f(x_1) = f(x_2) \iff d(f(x_1), f(x_2)) = 0 \iff d(x_1, x_2) = 0 \iff x_1 = x_2.$$

Moreover  $f$  is continuous via the  $\varepsilon$ - $\delta$  definition of continuity because for every  $\varepsilon > 0$  there is  $\delta = \varepsilon > 0$  such that

$$d_X(x_1, x_2) < \delta = \varepsilon \iff d_Y(f(x_1), f(x_2)) = d(x_1, x_2) < \varepsilon$$

and conversely  $f^{-1}: f(X) \rightarrow X$  is continuous because  $y_1, y_2 \in f(X)$  we get the relevant  $\varepsilon$ - $\delta$  bound by taking  $x_1 = f^{-1}(y_1)$  and  $x_2 = f^{-1}(y_2)$  in the displayed statement. Thus  $f$  is a homeomorphism between  $X$  and  $f(X)$ , which is what it means to be an embedding of  $X$  into  $Y$ .  $\square$

**Problem 5** (Munkres 21.9). Let  $f_n: \mathbb{R} \rightarrow \mathbb{R}$  be the function

$$f_n(x) = \frac{1}{n^3(x - 1/n)^2 + 1}.$$

Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be the zero function.

(a) Show that  $f_n(x) \rightarrow f(x)$  for each  $x \in \mathbb{R}$ .

(b) Show that  $f_n$  does not converge uniformly to  $f$ .

*Solution.* (a) For a given  $x$ , the sequence  $a_n = n^3(x - 1/n)^2 + 1 = n(nx - 1)^2 + 1$  has limit  $\infty$  as  $n$  goes to infinity because it is a nonconstant polynomial in  $n$  with positive leading coefficient. Therefore the sequence of its reciprocals  $1/a_n = f_n(x)$  converges to zero.

(b)  $f_n$  converges to  $f$  uniformly if for each  $\varepsilon > 0$  there is an  $n$  such that  $n > N$  implies that  $|f_n(x) - f(x)| = f_n(x) < \varepsilon$  for all  $x$ . However, given  $\varepsilon < 1$ , such an  $n$  doesn't exist, as for any  $n$ ,  $x = 1/n$  gives

$$|f_n(1/n) - f(1/n)| = f_n(1/n) = 1 > \varepsilon.$$

Therefore, the convergence of  $f_n$  to  $f$  is not uniform.  $\square$